

IIT Jodhpur	CSL-7480	Cryptography
Minor Exam	Semester 2, 2024-25	Instructor: Somitra Sanadhya
Time: 2 hours	2-4 pm, 21.02.25, LHC, Room 110	Max. Marks: 15

Note:

1. No queries will be entertained during the test.
2. If you do not understand any question then read it again.
3. Marks for each question/part are mentioned in the rightside margin. The total marks of each question are mentioned at the end of the question. A table showing the marks of all the question is provided at the end of the paper.

Notation: For n -bit strings x and y ,

- $x||y$ denotes concatenation of the two strings,
- $x \oplus y$ denotes bit-wise XOR of the two strings,
- $x + y$ means adding x and y and then taking remainder modulo 2^n (to ensure that the sum can be represented in n bits).

- (1) 1. Data compression is often used in data storage or transmission. Suppose you want to use data compression in conjunction with encryption. Does it make more sense to

- (a) Compress the data and then encrypt the result, or
- (b) Encrypt the data and then compress the result.

Justify your answer.

Total for Question 1: 1

- (1) 2. Your friend is learning Cryptography from you. You taught her the perfect security of the OTP scheme. She realizes that when the secret key is all-zeros, the encryption of the message m is m itself (since $m \oplus 0^n = m$, where m and c are n -bit long). She thinks that this is a bad idea. Hence, she makes a slight modification to the scheme, by discarding the all-zero key, but otherwise retaining the same scheme. More precisely, her modified OTP scheme is:

$$E(k, m) = \begin{cases} m \oplus k & \text{if } k \neq 0^n \\ m \oplus 1^n & \text{if } k = 0^n \end{cases}$$

Discuss whether her thinking is right. If yes, then why? If not, then reconcile it with the fact that $c = m$ when the secret key is all-zeros.

Total for Question 2: 1

- (1) 3. (a) Assume a user of a login system chooses a password which is either $abcd$ or $bedg$. Suppose the password is kept in an encrypted form on the server using the shift cipher, and an attacker gets to see the resulting ciphertext. Show how the attacker can determine the user's password, or explain why this is not possible.
- (1) (b) State Kerckhoff's principle. Why is it the right model to be used in Cryptography.
- (1) (c) Formally define perfect security. What is the primary limitation of perfect security?

Total for Question 3: 3

4. Explain how will the decryption be affected if the 3rd bit in the 2nd block of the ciphertext gets corrupted during transmission. Consider the following modes of operations:

- (1) (a) CBC mode
- (1) (b) Counter mode

Total for Question 4: 2

5. Let G_1 and G_2 be any secure PRG's from $\{0,1\}^n$ to $\{0,1\}^m$ where $m > n$. Are the following constructions secure PRG's? Why or why not? Justify your answer.
- (1) (a) $G'(s) = G_1(s) \oplus G_2(s)$
- (1) (b) $G'(s) = G_1(s) \parallel G_2(s \oplus 1)$

Total for Question 5: 2

6. Let P_k be a family of pseudorandom permutations indexed by the key k . That is, fixing a specific key k^* is equivalent to choosing a specific permutation $\pi_{k^*} \in P_k$ which is hard to distinguish from a random permutation.

For each of the following encryption schemes which encrypt $2n$ bit messages, state with justification whether the scheme is CPA secure? That is, either show an attack or prove that the scheme is secure. The $\parallel$ symbol stands for concatenating two strings.

- (1) (a) $E(k, m_1 \parallel m_2) = \pi_k(m_1) \parallel \pi_k(m_2)$
- (1) (b) $E(k, m_1 \parallel m_2) = IV \parallel (\pi_k(IV + 1) \oplus m_1) \parallel (\pi_k(IV + 2) \oplus m_2)$
where IV is a random n bit string (and $\oplus$ and $+$ are as defined in the notation earlier).

Total for Question 6: 2

- (1) 7. (a) Prove that a 2-round Feistel construction is not a PRP. (i.e. show an attack which breaks the PRP property).
- (2) (b) Assume that you have a block cipher E which supports an n -bit key and has t -bit block size. To increase the key-size to $2n$ bits, your friend suggests using the following encryption algorithm:

$$E'(k_1 \parallel k_2, m) = E(k_1, E(k_2, m)).$$

Keys k_1 and k_2 are n bits long, and the message m is of length t bits. Show an attack against the encryption scheme above. Mention the time and space complexity of your attack in terms of n and t .

Total for Question 7: 3

- (1) 8. Derive the "birthday bound" for Cryptographic hash functions. What is its impact on the design of such functions?

Total for Question 8: 1

Table of marks:

Question:	1	2	3	4	5	6	7	8	Total
Points:	1	1	3	2	2	2	3	1	15
Score:									

Write briefly. Best wishes.